

Serotonin Syndrome



Section I: Scenario Demographics

Scenario Title:	Serotonin Syndrome
Date of Development:	(19/02/2016)
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	Expose learners to a hyperthermic, altered patient with serotonin syndrome.
CRM Objectives:	1. Demonstrates problem solving through prompt ABC assessment and implementation of concurrent management approach 2. Demonstrates situational awareness by avoiding fixation error and reassessing the situation as the case progresses
Medical Objectives:	1. Initiates broad initial workup for the febrile altered patient 2. Recognizes the serotonergic toxidrome and initiates rapid cooling and medical therapy 3. Employs a tailored approach to ventricular dysrhythmia in the patient with serotonin syndrome

Case Summary: Brief Summary of Case Progression and Major Events
A 27-year-old female presents hot and altered to the ED with EMS. Likely cause is serotonin syndrome, precipitated by being on citalopram and methadone in the setting of a recent cocaine binge (all increase serotonin levels). She will develop Torsades de Pointes as a complication which must be treated with MgSO4. She will become increasingly agitated and febrile, requiring IV benzodiazepines, active cooling, and consideration of intubation with paralysis to achieve normothermia.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.
Buckley NA, Dawson AH, Isbister GK. Serotonin syndrome. BMJ. 2014 Feb 19;348:g1626
Isbister GK, Buckley NA, Whyte IM. Serotonin toxicity: a practical approach to diagnosis and treatment. Medical Journal of Australia. 2007 Sep 17;187(6):361.



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
<i>Select most important dimension(s)</i>			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☒ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☒ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: Fans	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Female wig. Diaphoresis.			
E. Approximate Timing			
Set-Up:	10 min	Scenario:	15 min
		Debriefing:	30 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 27-year-old female was found by her boyfriend this morning seeming confused and warm. He called EMS. She has a history of opioid abuse and is on methadone, but he swears that she has taken this as prescribed and has not done any prescription pain meds lately. They did "party a lot yesterday", but she was otherwise well, with no complaints of fever before today. With EMS the patient was noted to be diaphoretic, febrile and quite agitated. She has been placed in a resuscitation bay.

B. Patient Profile and History

Patient Name: Sandy Jones		Age: 27		Weight: 60kg	
Gender: ☐ M ☒ F		Code Status: Full			
Chief Complaint: Fever and Altered LOC					
History of Presenting Illness: As above					
Past Medical History:	Depression		Medications:	Citalopram 40mg daily	
	Opioid Abuse			Methadone	
Allergies: Penicillin					
Social History: Binge EtOH on weekends, 1ppd smoker, + cocaine over weekend at a party					
Family History: Unknown					
Review of Systems:	CNS:	Confused this morning per boyfriend. Not oriented with EMS			
	HEENT:	No history of head injury/ recent trauma			
	CVS:	Nil			
	RESP:	No recent cough.			
	GI:	No recent abdo pain, vomiting, or diarrhea			
	GU:	No dysuria or recent urinary symptoms			
	MSK:	Nil	INT:	Nil	

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 110 /min		BP: 122/75		RR: 18/min	
Rhythm: NSR		T: 39.4°C		Glucose: 6.4mmol/L	
				O ₂ SAT: 99%	
				GCS: 14 (E4 V4 M6)	
General Status: Shivering, diaphoretic					
CNS:	Not oriented. Agitated. Seems "jumpy"				
HEENT:	No signs HI. Pupils 4mm reactive bilaterally				
CVS:	Normal				
RESP:	Normal				
ABDO:	Normal				
GU:	Normal				
MSK:	Inducible clonus, hyper-reflexia		SKIN:	Sweaty	

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Wide QRS HR: 110/min BP: 120/80 RR: 18/min O ₂ SAT: 99% T: 39.5°C	GCS 14 as above. ++ agitated Inducible clonus + hyper-reflexia	<u>Learner Actions</u> - ☐ IV, O ₂ , monitors - ☐ Blood work (tox + sepsis + TSH + CK) - ☐ Cap glucose - ☐ ECG (wide QRS) - ☐ NaHCO ₃ for wide QRS - ☐ IV NS 1L bolus - ☐ Consider antipyretic - ☐ Broad-spectrum antibiotics - ☐ Urine dip, βHCG - ☐ ± Portable CXR - ☐ Consider head CT	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Antipyretic → no effect - 1L IV NS → HR 105, BP 125/85 - 1amp NaHCO ₃ → narrowing of QRS (ECG with prolonged QT) <u>Triggers</u> <i>For progression to next state</i> - All actions complete or 5 minutes → 2. Torsades
2. Torsades Rhythm: torsades HR: 180/min BP: 60/30 RR: 18/min O ₂ SAT: 93% T: 40.5°C	Patient GSS drops to 8 (moaning)	<u>Learner Actions</u> - ☐ Cardioversion at 100-200J - ☐ MgSO ₄ 2g IV - ☐ Repeat ECG	<u>Modifiers</u> - Cardioversion before MgSO ₄ → brief NSR then back to torsades <u>Triggers</u> - MgSO ₄ given → 4. More Agitated - No MgSO ₄ by 3 min → 3. Arrest
3. Arrest Rhythm: torsades HR: 180/min BP: 0/0 RR: 0 O ₂ - ???	Patient loses pulse	<u>Learner Actions</u> - ☐ IV MgSO ₄ - ☐ Defibrillation at 200J - ☐ High quality CPR	<u>Modifiers</u> - Defibrillation only → No ROSC <u>Triggers</u> - MgSO ₄ given → 4. More Agitated
4. More Agitated Rhythm: NSR HR: 120/min BP: 140/90 RR: 16/min O ₂ SAT: 99% T: 40.5	Patient GCS 13. Very jumpy. Speaking nonsense.	<u>Learner Actions</u> - ☐ IV Benzos until more calm - ☐ Active cooling - ☐ Consider intubation for adequate sedation & cooling - ☐ Call Poison Control	<u>Modifiers</u> - 1 st dose benzos → HR 110 <u>Triggers</u> - Cooling + benzos given → 5. Labs Back - 4 min → 5. Labs Back
5. Labs Back Rhythm: NSR HR: 100/min BP: 130/80 RR: 16/min O ₂ SAT: 99%	Give lab results at beginning of state.	<u>Learner Actions</u> - ☐ Consider cyproheptadine - ☐ Continue active cooling - ☐ ICU consult	<u>END CASE PRN</u>

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 138	K: 4.5	Cl: 102	HCO ₃ : 22	BUN: 6	Cr: 70	Glu: 6.8
VBG	pH: 7.28	PCO ₂ : 30	PO ₂ : 100	HCO ₃ : 18	Lactate: 3.4	
TSH: 0.4 (normal)		CK: 100 (normal)				
WBC: 14		Hg: 122		Hct: 0.4		Plt: 300
ASA: <0.36		Tylenol <0.15		EtOH: <2		Osm: 300 (OG=3)

Images (ECGs, CXRs, etc.)

CXR – normal



CXR source:

<https://radiopaedia.org/cases/normal-chest-radiograph-female-1>

Post-intubation CXR – normal



CXR source:

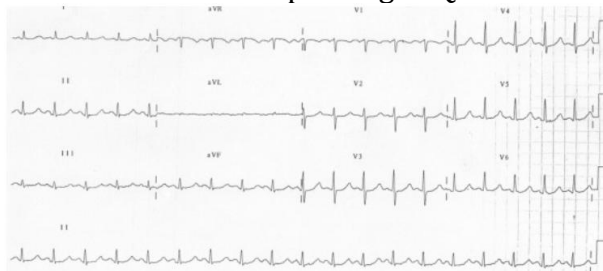
<https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg>

1st ECG = prolonged QRS + QT



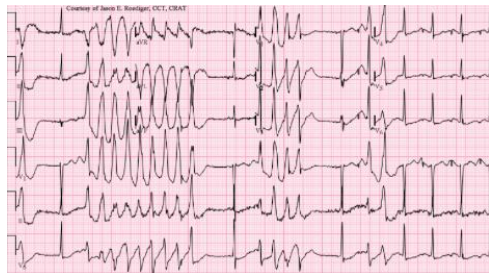
ECG source: <https://lifeinthefastlane.com/ecg-library/basics/tca-overdose/>

ECG after NaHCO₃ = prolonged QT



ECG source: https://lifeinthefastlane.com/ecg-library/basics/qt_interval/

3rd = Torsades



ECG source: https://en.wikipedia.org/wiki/Torsades_de_pointes



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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What was your initial approach to the febrile altered patient? 2) When did you consider serotonin syndrome as a possible diagnosis? What clued you in to this diagnosis? 3) This patient did not take an overdose of her medications. Why might she have developed serotonin syndrome anyways? 4) What are the main clinical features of serotonin syndrome? 5) What is the main priority in the management of a patient with serotonin syndrome? What are some end-stage considerations for management? 6) Why did the patient have a prolonged QRS? Was this part of the serotonin syndrome or related to something else? (Answer: cocaine-induced Na-channel toxicity)	
Key Moments	
1) Identification of sick patient who is febrile and altered with initially broad work-up	
2) Recognition of serotonin syndrome and treatment of same with benzos and active cooling	
3) Recognition of torsades and appropriate, rapid treatment	